# PahaarPulse — Gemma Local Vision Inference

Technical Specification, Open-Weight Model Architecture & Verification Report • October 2026

***Executive Summary:*** *This technical specification details the enhancement of the PahaarPulse Landslide Risk Management System with an on-premises open-weight Vision-Language Model (VLM) provider powered by* ***Google Gemma 3 4B (gemma3:4b)*** *via the Ollama inference runtime. The architecture provides complete offline hazard screening during regional communication outages while strictly preserving the hosted Google AI provider (gemini-3.5-flash), Pydantic schemas, database integrity, and geotechnical safety guardrails.*

## 1. System Architecture & Component Interaction

PahaarPulse utilizes an observational AI triage pipeline to screen ground field reports submitted by mobile and web clients. The architecture integrates a provider-agnostic abstraction layer with strict atomicity and idempotency guarantees.

## 2. Open-Weight Vision Model Selection

Candidate vision-language models were benchmarked against the host machine's hardware profile (NVIDIA GeForce RTX 4060 Laptop GPU with 8,188 MiB VRAM, 16 GB RAM, 269 GB free disk space):

| Model | Publisher | Params | Quant. Size | VRAM Req. | License | Assessment |
|---|---|---|---|---|---|---|
| **gemma3:4b** | Google DeepMind | 4 Billion | ~3.1 GB | ~3.8 GB | Gemma Terms | **Selected Primary:** Native vision tokens; fits in 8 GB VRAM with 4 GB headroom. |
| llama3.2-vision:11b | Meta AI | 11 Billion | ~7.9 GB | ~8.8 GB | Llama 3.2 Comm. | Exceeds 8 GB threshold; requires CPU offloading, raising latency. |
| moondream:1.8b | Vikhyat | 1.8 Billion | ~1.7 GB | ~2.2 GB | Apache 2.0 | Compact but exhibits higher hallucination on geotechnical schemas. |

## 3. Evidence-Grounded Geotechnical Screening Rules

The system prompt enforces four non-negotiable safety guardrails:

- **Separation of Evidence:** Features in detected_hazards must be directly visible in the photograph. Observer notes are treated as unverified claims and must not be reported as visual confirmation.
- **Conservative Triage & Indeterminate State:** If image quality, lighting, perspective, distance, or vegetation obscures the slope, the classification defaults to indeterminate. Uncertain reports are never defaulted to low.
- **No Safety Clearances:** The model is strictly prohibited from declaring an area 'safe' or 'cleared'. Outputs represent preliminary triage and do not substitute for on-site geotechnical engineering inspections.
- **Uncalibrated Confidence:** Confidence scores reflect model self-assessed clarity (0.0 to 1.0), not empirical statistical probabilities.

## 4. Administrative Security & Fail-Closed Policy

The review (PATCH /api/field-reports/{id}/review) and re-analysis (POST /api/field-reports/{id}/analyze) endpoints are guarded by verify_coordinator_authorization:

- **Production Fail-Closed:** In production environments (ENVIRONMENT=production), if COORDINATOR_API_KEY is not configured, endpoints immediately reject all requests with HTTP 503 Service Unavailable, preventing accidental public exposure.
- **Credential Validation:** When configured, endpoints enforce credential checks via X-Coordinator-Key header or Authorization: Bearer <token>.
- **Development Ergonomics:** In local development environments without configured keys, access is permitted with logged security warnings.

## 5. Test Verification & Real Model Smoke Test

Verification was performed across automated test suites, real multimodal inference, and database preservation checks:

| Verification Scope | Test Target / Suite | Pass Count | Duration | Status |
|---|---|---|---|---|
| Backend Automated Suite | python -m unittest discover backend/tests | 172 / 172 | 1.65s | **PASS (100%)** |
| Public Image Acquisition Suite | backend/tests/test_public_image_discovery.py | 20 / 20 | 0.048s | **PASS (100%)** |
| API Authorization Suite | backend/tests/test_api_authorization.py | 25 / 25 | 0.45s | **PASS (100%)** |
| Local Vision & Security Provider | backend/tests/test_local_vision_provider.py | 15 / 15 | 0.26s | **PASS (100%)** |

| | | | | |
|---|---|---|---|---|
| VLM Benchmark & Eval Suite | backend/tests/test_vlm_evaluation.py | 22 / 22 | 0.025s | **PASS (100%)** |
| Mobile Unit Suite | node --test src/__tests__/run-mobile-tests.mjs | 17 / 17 | 6.63ms | **PASS (100%)** |
| Next.js Production Build | npm run build (Next.js 16 App Router) | Clean Build | 1.26s | **CLEAN (24 Routes)** |
| Mobile Static Typing | npx tsc --noEmit (Expo React Native) | 0 Errors | — | **CLEAN** |
| Real Hosted Vision Inference | scratch/real_smoke_test.py (gemini-3.5-flash) | Disposable E2E | 10.56s | **VERIFIED LIVE** |
| Real Local Gemma 3 VLM Inference | scratch/test_local_e2e_workflow.py (ollama:gemma3:4b) | Disposable E2E | 2.98s (warm) | **VERIFIED LIVE** |
| Database Preservation | PostgreSQL operational tables (11 public tables) | Zero Data Loss | — | **PRESERVED** |

## 6. Security Audit Findings & Hardening Remediation

| Ref | Component | Security Finding / Impact | Severity | Remediation Status |
|---|---|---|---|---|
| SEC-01 | deps.py | String comparison vulnerable to timing side-channels against coordinator secret. | Medium | **FIXED:** secrets.compare_digest |
| SEC-02 | deps.py | Container ENV=production alias not detected alongside ENVIRONMENT. | Low | **FIXED:** Added ENV; fail-closed verified |
| SEC-03 | fieldReports.ts & BFF | NEXT_PUBLIC_COORDINATOR_API_KEY exposed in browser client bundles. | High | **FIXED:** Zero client keys; Next.js server BFF |
| SEC-04 | deps.py | Potential credential leakage in server exception details or logs. | High | **VERIFIED SAFE:** No keys in output |
| SEC-05 | field_reports.py | Public routes must remain open for field reporters and offline sync. | Low | **VERIFIED:** Upload & sync open |
| SEC-06 | coordination & alerts | 16 coordinator mutation routes and 2 audit read endpoints lacked auth. | Critical | **FIXED:** verify_coordinator_authorization (18 routes) |
| SEC-07 | Next.js Proxy BFF | Proxy risked acting as confused deputy for anonymous web visitors. | High | **FIXED:** Fail-closed in prod (HTTP 401); Origin checked |

## 7. Scientific Validation Phase 1 & Benchmark Dataset Compatibility

**Repository Ground-Truth Audit:** An exhaustive audit confirms that **no expert-adjudicated, ground-truth labelled landslide image dataset currently exists in the repository**. In accordance with geotechnical and AI ethics, zero accuracy metrics are fabricated on unadjudicated data. Phase 1 establishes a reproducible, scientifically defensible evaluation pipeline:

- **Target Triad Separation:** Evaluates **Target 1** (observable surface indicators: tension cracks, scarps, blockages) and **Target 2** (operational triage urgency: low to critical, defaulting to indeterminate when obscured). **Target 3** (future failure probability) is out of scope for field photos alone as it requires subsurface boreholes, DEM slopes, and pore pressures; model confidence is visual certainty, not calibrated failure probability.
- **Dual-Expert Ground-Truth Protocol:** Formalized in GROUND_TRUTH_PROTOCOL.md with independent blinding, Cohen's Kappa threshold (kappa ≥ 0.70) to qualify datasets, consensus adjudication, 4 quality tiers, and 4 hard exclusion reasons.
- **18-Column Manifest & Leakage Guards:** Defined in field_photo_benchmark_template.csv with cryptographic SHA-256 byte digests, train/dev/test split allocations, cross-split duplicate rejection, and Wilson score 95% confidence intervals.

External candidate benchmark datasets were audited against licensing, modality, and annotation compatibility:

| Dataset ID | Source & Citation | Modality | Annotation | License Status | Suitability Assessment |
|---|---|---|---|---|---|
| **HLAD** | Zenodo 10.5281/zenodo.22811448 | RGB UAV/Drone (Himalayas) | Pixel masks & bounding boxes | BLANK / UNVERIFIED | **BLOCKED:** License missing on Zenodo; spatial masks incompatible with triage. |
| **Landslide4Sense** | IEEE GRSS / HPS Data Fusion | 14-band satellite rasters (128x128) | Binary masks (0/1 background/scar) | Academic Non-Comm. | **BLOCKED:** Incompatible raster modality; binary masks orthogonal to VLM photo screening. |
| **PahaarPulse-Benchmark** | Internal operational field repo | Oblique RGB photos (.jpg, .png) | 5-tier triage + 8 indicators | Internal / Apache-2.0 | **TARGET SCHEMA:** Fully compatible; pending independent expert review. |

**Strict Input Rejection Barriers:** backend/app/services/vlm_evaluation.py enforces typed guardrails: UnverifiedLicenseError (blocks unverified licenses), UnsupportedModalityError (blocks multi-spectral/satellite rasters: .tif, .npy, .h5), and MismatchedAnnotationSchemaError (blocks binary masks and bounding boxes lacking valid categorical hazard severity labels).

## 8. Status Verification Matrix

| Capability / Requirement | Implemented | Automated Tested | Real Model Tested | Security Verified | PDF Verified |
|---|---|---|---|---|---|
| Hosted Google GenAI Vision (gemini-3.5-flash) | YES | YES | YES (Live E2E) | YES | YES |
| Local Ollama Vision Provider (gemma3:4b) | YES | YES (Contract) | YES (Live: 2.98s, 3.8GB) | YES | YES |
| Evidence-Grounded Triage Prompt Rules | YES | YES | YES (Live E2E) | YES | YES |
| Provider Fallback & Provenance Tracking | YES | YES | YES (Unit/Integ) | YES | YES |
| Production Fail-Closed Coordinator Guard | YES | YES (25 Tests) | YES (18 Endpoints) | YES (Fail-Closed) | YES |
| VLM Evaluation Harness & Provenance Guard | YES | YES (22 Tests) | YES (Harness Validated) | YES | YES |
| Phase 1 Field Benchmark & Leakage Guards | YES | YES (22 Tests) | YES (CLI Tested) | YES | YES |
| Public Image Discovery & Licensing | YES | YES (20 Tests) | YES (6 Verified) | YES (Magic-Bytes) | YES |

## 9. Operational Guide & Remaining Production Blockers

**User Session & Reviewer Identity Prerequisite:** The application currently lacks an interactive login screen, session cookie/JWT manager, and user-level RBAC. Field report reviews record reviewed_by = None to prevent fabricating reviewer identity or attributing actions to a shared key. Full production readiness requires deploying a dedicated session/identity layer prior to multi-operator deployment.

**Validation & Calibration Blockers:** The local VLM architecture is functional and tested, but cannot be considered clinically or scientifically validated until tested against an independent, expert-annotated benchmark of regional landslide photographs. Confidence scores remain uncalibrated model-reported values.

```
# Step 1: Install Ollama on Windows
winget install Ollama.Ollama

# Step 2: Start daemon & pull Gemma 3 4B Vision model (~3.1 GB)
ollama serve
ollama pull gemma3:4b

# Step 3: Switch provider in backend/.env
GEMMA_PROVIDER=ollama
OLLAMA_BASE_URL=http://localhost:11434
OLLAMA_MODEL=gemma3:4b
OLLAMA_TIMEOUT=60.0

# Step 4: Launch unified backend server
python -m uvicorn backend.app.main:app --host 0.0.0.0 --port 8000 --reload
```